Procurement: the track unit on Nhien’s Rev B frame

Date 16 Oct 2026, updated 19 Oct (3 x 5 rail screws, stage-2 pins 2.5 x 16). Prices S$ ex GST as seen in listings; confirm at checkout. Rev B = Nhien’s pack (`RB4107 Rev B`), whose BOM covers everything in section 2. The track comes in two options, A and B; everything else is the same whichever is chosen, so section 2 and the pulleys can be ordered first.

1A. Track, option A: my printed track

Part	Qty	Make / buy	Where	Price	Arrives
Track, TPU 95A, `stl/track_TPU_x4.stl` (drawing TS-01)	4	**Make**: Fab Lab UP300 (TPU head)	Element14 SG TPU 95A filament if the lab has none	~S$45/kg, uses 130 g	filament 1-2 days; 3 h print each
Fit-test slice, `stl/fit_test_track_5mm_TPU.stl`	1	Make	same	-	10 min
Stand-in pulley, `stl/stand_in_pulley_20T_PLA_test_only.stl`	1	Make, PLA	any printer	-	1 h

Deck: no change needed; the 3 mm cleats sit 5 mm under the deck.

1B. Track, option B: Nhien’s bought belt with cleats (Rev B B10)

Part	Qty	Make / buy	Where	Price	Arrives
Track belt HTD 550-5M-15, 110 teeth, 15 wide (e.g. SKF PHG 550-5M-15), drawing TS-01B	4	**Buy**	RS SG, stock no. 058-6535 or MISUMI SG HTD 5M belts	see listing (about S$25-35 each)	RS SG stock: 1-3 days
Cleats, rubber 10 x 7.9 x 15, bonded on the belt back every 5th tooth (25 mm), 22 per belt	88	**Buy + bond**: belt shop	Local belt / conveyor shop (Fenner, Megadyne or Gates distributor), with the belts	quote	allow 1-2 weeks
Deck change for the belt: notch P12 13 x 135 over each front lane (x -25 to 110), or cut the cleats to 5 high	1	**Make**: with the deck laser cut, or file / mill after	Professional Workshop	-	with the deck

1C. Wheels, both options

Part	Qty	Make / buy	Where	Price	Arrives
Timing pulley HTD 5M 20T, 15 mm belt, flanged (19 Oct: MISUMI economy HTD 5M, code **H5M**, not S5M). **Hinge wheels: 4 + 1 with a finished Ø8 bore**: shaft-hole type H (plain round, no tapped hole) if the configurator offers it for 20T, else P (round with tapped hole, ignore the screw; pin hole 90° from it). **Idlers: 4 with the smallest bore offered**, bored to Ø16 at the workshop (20T stops at about Ø15, so no 16 bore). Before ordering, check on the drawing: width over the flanges 19 or less (the lane is 20) and no hub (shape B has a Ø19 hub)	8 + 1	**Buy**	MISUMI SG, Timing Pulleys HTD 5M	~S$20 each	ships 1 day from SG stock; allow 2-5 days
Pulley bore modification: 4 x bore Ø16 H7 (idlers, M2) only; the hinge wheels come bored Ø8. Stage-2 pulleys M3 (Nhien's): ream Ø8 H7. Ø2.5 cross holes drilled on our drill press in a V-block, shaft drilled through them at assembly (TS-06, TS-11)	4 + 2	**Make**: Professional Workshop request	E3-01-06	free	workshop queue, allow 1-2 weeks
Spring pins ISO 8752 2.5 x 24 (hinge pulleys, replaces E-clips F22 and the D-flats)	4 + 4 spare	**Buy**	RS SG spring pins	~S$5 / pack	RS SG stock
Spring pins ISO 8752 2.5 x 16 (stage-2 pulleys M3 only, if Nhien pins them: the pulley is Ø17.5, so 24 is too long)	2 + 2 spare	**Buy**	RS SG spring pins	~S$5 / pack	RS SG stock

Part	Qty	Make / buy	Where	Price	Arrives
Set screws M4 x 5 cone point DIN 914 (stage-2 pulleys M3 only, if Nhien keeps his pulley's tapped hole: cone point into a seat spot-drilled in the shaft through that hole, Loctite 243; the hinge wheels have no hub in the 20 mm lane, so they stay pinned; plan Appendix C item 17)	2 + 2 spare	**Buy**	RS SG set screws	~S$3 / pack	RS SG stock
Hinge shaft: plain round Ø8 h7 ground steel, 300 mm (100 mm to practise on) (replaces the D-shaft B13; cut to 82.3 and 77.3, no grooves)	1	**Buy**, cut	RS SG linear shafts or MISUMI SG PSFJ8	~S$10	RS SG stock

1D. Tension and rail support (from the design review), both options

Part	Qty	Make / buy	Where	Price	Arrives
Tensioner tab, Al angle 20 x 20 x 3 cut 30 with a 7.5 x 8 ear, M4 tapped (TS-09)	4	**Make**: hacksaw, drill, M4 tap	from BOM B16 angle (order 300 mm instead of 100)	-	1 h
Tensioner slider, an 8 mm slice of the same angle, leg 19 with Ø6.4, foot 8 (TS-09; replaces the M6 washer under the nut)	4	**Make**: hacksaw, drill	from the same angle (order 400 mm)	-	30 min
Thread-forming screws 3 x 5 pan head for plastics (rail to carrier angle, 3 per rail; 3 x 5 not 3 x 10: the rail is 5 thick)	24 + 6 spare	**Buy**	RS SG thread-forming screws	~S$6 / pack	RS SG stock
M3 x 6 SHCS ISO 4762 (carrier angles to plates 24, tensioner tab ears 4)	28 + 4 spare	**Buy**	RS SG	~S$5 / pack	RS SG stock
Jack screw M4 x 20 socket head + M3 x 6 for the tab	4 + 4	**Buy**	RS / local	-	stock
New M3 tapped holes in P1 and P2: (x +-223, y +11) for the tab (TS-09) and six at (x +-40, +-105, +-170; y +-3.5) for the carriers (TS-10)	4	**Make**: add to the DXF, or drill 2.5 + tap M3 on site	with the plates	-	with the plates
Rail carrier angle, Al 10 x 10 x 1.5 x 186, plate leg cut to 6.5, 3 + 3 holes (TS-10; replaces the 32 RS 427-991 clips)	8	**Make**: from 1.6 m of angle	local metal supplier / MISUMI SG	~S$10	1-3 days

Part	Qty	Make / buy	Where	Price	Arrives
Rails P3-P6 trimmed from 189 to 186 at the idler end (TS-10, TS-11)	8	Make: hacksaw	-	-	10 min

2. Nhien’s (frame, drive, hinge, electronics), from the Rev B BOM

Part	Qty	Make / buy	Where
Side plates P1 (front) and P2 (rear), 5 mm Al, laser + tap	2 + 2	Make	DXF in the pack; Professional Workshop or a laser shop on the vendor list
Slide rails P3-P6, UHMW strip 189 x 20 x 5	8	Buy + drill	MISUMI SG UPA-189-20-5
Rail brackets RS PRO 427-991 (angle 15 x 14.6 x 10)	32	Buy	RS SG stock no. 427-991
Hinge shafts P20 A 82.3 + B 77.3, now plain round Ø8 h7 (section 1C), no D-flat, no E-clip grooves	2	Buy stock, cut	see 1C
Sector gear P11: proposed KHK SSAY1-100 (module 1, 100 teeth, 20°, face 6, bore 10, S45C, flat plate) cut to a 23-tooth sector (plan 5.14); needs Nhien’s OK on the centre distance 60.1 to 60.2 (P13 slot 1 mm closer) or a 22-tooth module 1 pinion in place of M6	1	Buy, cut	KHK catalog (thin-face hubless SSAY series, about US$43) or MISUMI SG (KHK brand), search SSAY1-100; plan B any 1M 100T 45# steel gear from Shopee / AliExpress, faced to 6 mm at the Professional Workshop
Bearing 688-2RS 8 x 16 x 5	14	Buy	RS SG ball bearings
Shoulder screw ISO 7379 Ø8 x 25 M6 + washers, shim, spacer, nyloc	4 sets	Buy	RS SG shoulder screws
2020 profile x 110 (P7) x2, 3030 x 160 (P8) x2, corner posts P15 x4	1 m + 0.5 m	Buy, cut	MISUMI SG / local
Corner brackets: standard cast 2020 and 3030 inside corner brackets instead of the P18 / P19 angle offcuts (bigger contact area, design review); T-nuts, shims 8x11x0.5 and 8x11x1; E-clips F22 no longer needed	-	Buy	MISUMI SG / local
Deck P12 (slot H 3 mm longer, see pack page 2), tray P16, cover P22, motors, drivers, ESP32, battery	-	Theirs	Rev B BOM
Bearing seats in P1, P2, P17: laser-cut Ø15.8, reamed Ø16 H7 on site, 688-2RS retained with Loctite 641 (TS-11)	6	Make	Professional Workshop, 16 H7 reamer

3. Choosing between A and B

Option A is in-house: four 3 h prints, no supplier, 5 mm deck clearance, and the cleat spacing is matched to the ES1 step. Option B is bought: belts from RS in days, but the 88 cleats need a belt shop (1-2 weeks) and the deck needs a notch or shorter cleats. Wheels, bore modification, plates, shafts, bearings and shoulder screws are the same in both.

4. Order of events

1. Order the 8 plain-bore pulleys today (critical path, then the workshop bore mod); for option B also order the 4 belts and send them with TS-01B to the belt shop. For option A print the stand-in pulley and fit-test slice, then the 4 tracks as soon as a UP300 slot is free.
2. Nhien: laser plates and deck (slot H 74 long; notched if option B), order rails, bearings, shoulder screws, profiles.
3. Workshop: pulley bores, plate tapping, shaft cutting (one request form).
4. Assemble per Rev B sheet 3; fit the tracks on the real pulleys; tension at the slots.